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SE Lab – 7

Step – 1

```
PS C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7> py -m pip install hypothesis pytest
Requirement already satisfied: hypothesis in c:\users\nikhi\appdata\local\programs\python\python314\lib\site-packages (6.170.15)
Requirement already satisfied: pytest in c:\users\nikhi\appdata\local\programs\python\python314\lib\site-packages (9.2.0)
Requirement already satisfied: sortedcontainers<3.0.0,>=2.1.0 in c:\users\nikhi\appdata\local\programs\python\python314\lib\site-packages (from hypothesis) (2.4.0)
Requirement already satisfied: colorama>=0.4 in c:\users\nikhi\appdata\roaming\python\python314\site-packages (from pytest) (0.4.6)
Requirement already satisfied: iniconfig>=2 in c:\users\nikhi\appdata\local\programs\python\python314\lib\site-packages (from pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in c:\users\nikhi\appdata\roaming\python\python314\site-packages (from pytest) (25.0)
Requirement already satisfied: pluggy<2,>=1.6 in c:\users\nikhi\appdata\local\programs\python\python314\lib\site-packages (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in c:\users\nikhi\appdata\roaming\python\python314\site-packages (from pytest) (2.19.2)
```

Step – 2

```
PS C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7> py processor.py
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): DT-UP-3
Route DT->UP: 3 Zones, Fare $4.75

==== Running lookup_concession_discount ====
Enter concession category (e.g. STUDENT, SENIOR): STUDENT
Multiplier: 0.5

==== Running compute_passenger_density ====
Enter passengers/km (e.g. 50/10.0): 50/10.0
Density: 5.0
```

Step – 4

```
PS C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7>  pytest test_processor.py
============================================================================= test session starts ==============================================================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7
plugins: anyio-4.14.2, hypothesis-6.170.15, langsmith-0.14.9, platformdirs-4.14.4, typeguard-4.6.0
collected 3 items

test_processor.py FFF                                                                                                                                                  [100%]

================================================================================== FAILURES ==================================================================================
_____________________________________________________________________ test_parse_transit_corridor_no_crash _____________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 6, in test_parse_transit_corridor_no_crash
  |     def test_parse_transit_corridor_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 9, in test_parse_transit_corridor_no_crash
    |     parse_transit_corridor(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 2, in parse_transit_corridor
    |     orig, dest, zones_str = corridor_code.strip().split('-')
    |     ^^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 3, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 11, in test_parse_transit_corridor_no_crash
    |     assert False, f"parse_transit_corridor crashed with: {e}"
    | AssertionError: parse_transit_corridor crashed with: not enough values to unpack (expected 3, got 1)
    | assert False
    | Failing test case: test_parse_transit_corridor_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 9, in test_parse_transit_corridor_no_crash
    |     parse_transit_corridor(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 2, in parse_transit_corridor
    |     orig, dest, zones_str = corridor_code.strip().split('-')
    |                             ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 11, in test_parse_transit_corridor_no_crash
    |     assert False, f"parse_transit_corridor crashed with: {e}"
    | AssertionError: parse_transit_corridor crashed with: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_parse_transit_corridor_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________________________________ test_lookup_concession_discount_safe _____________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 15, in test_lookup_concession_discount_safe
  |     def test_lookup_concession_discount_safe(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 18, in test_lookup_concession_discount_safe
    |     lookup_concession_discount(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 16, in lookup_concession_discount
    |     return discounts[category]
    |            ~~~~~~~~~^^^^^^^^^^
```

```
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 20, in test_lookup_concession_discount_safe
    |     assert False, f"lookup_concession_discount crashed with: {e}"
    | AssertionError: lookup_concession_discount crashed with: ''
    | assert False
    | Failing test case: test_lookup_concession_discount_safe(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py:19
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 18, in test_lookup_concession_discount_safe
    |     lookup_concession_discount(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 8, in lookup_concession_discount
    |     category = category_name.strip().upper()
    |                ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 20, in test_lookup_concession_discount_safe
    |     assert False, f"lookup_concession_discount crashed with: {e}"
    | AssertionError: lookup_concession_discount crashed with: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_lookup_concession_discount_safe(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py:19
    +------------------------------------
_____________________________________________________________________ test_compute_passenger_density_safe _____________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 24, in test_compute_passenger_density_safe
  |     def test_compute_passenger_density_safe(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 27, in test_compute_passenger_density_safe
    |     compute_passenger_density(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 20, in compute_passenger_density
    |     passengers = float(passengers_str)
    |                  ^^^^^^^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 29, in test_compute_passenger_density_safe
    |     assert False, f"compute_passenger_density crashed with: {e}"
    | AssertionError: compute_passenger_density crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_compute_passenger_density_safe(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 27, in test_compute_passenger_density_safe
```

```
    | )
    +------------------ 2 -------------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 27, in test_compute_passenger_density_safe
    |     compute_passenger_density(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 19, in compute_passenger_density
    |     passengers_str, km_str = occupancy_ratio_str.split('/')
    |     ^^^^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 29, in test_compute_passenger_density_safe
    |     assert False, f"compute_passenger_density crashed with: {e}"
    | AssertionError: compute_passenger_density crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_compute_passenger_density_safe(
    |     s='',  # or any other generated value
    | )
    +------------------ 3 -------------------
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 27, in test_compute_passenger_density_safe
    |     compute_passenger_density(s)
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\processor.py", line 19, in compute_passenger_density
    |     passengers_str, km_str = occupancy_ratio_str.split('/')
    |                              ^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7\test_processor.py", line 29, in test_compute_passenger_density_safe
    |     assert False, f"compute_passenger_density crashed with: {e}"
    | AssertionError: compute_passenger_density crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_compute_passenger_density_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
============================================================================ warnings summary ============================================================================
..\..\..\..\..\..\Users\nikhi\AppData\Local\Programs\Python\Python311\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\nikhi\AppData\Local\Programs\Python\Python311\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================================================== short test summary info ===========================================================================
FAILED test_processor.py::test_parse_transit_corridor_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_lookup_concession_discount_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_compute_passenger_density_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================================================ 3 failed, 1 warning in 1.92s ============================================================================
```

Step – 5

```
PS C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7> py processor.py
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): DT-UP-3
Route DT->UP: 3 Zones, Fare $4.75

==== Running lookup_concession_discount ====
Enter concession category (e.g. STUDENT, SENIOR): STUDENT
Multiplier: 0.5

==== Running compute_passenger_density ====
Enter passengers/km (e.g. 50/10.0): 50/10.0
Density: 5.0
```

Step – 6

```
PS C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7> py processor.py
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): DT-UP-3
Route DT->UP: 3 Zones, Fare $4.75

==== Running lookup_concession_discount ====
Enter concession category (e.g. STUDENT, SENIOR): STUDENT
Multiplier: 0.5

==== Running compute_passenger_density ====
Enter passengers/km (e.g. 50/10.0): 50/10.0
Density: 5.0
PS C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7> pytest test_processor.py
============================================================================ test session starts =============================================================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Personal\PES College Related\Sem - 5\SE\Lab\Lab - 7
plugins: anyio-4.14.2, hypothesis-6.170.15, langsmith-0.14.9, platformdirs-4.14.4, typeguard-4.6.0
collected 3 items

test_processor.py ...                                                                                                                                                  [100%]

============================================================================ warnings summary ============================================================================
..\..\..\..\..\..\Users\nikhi\AppData\Local\Programs\Python\Python311\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\nikhi\AppData\Local\Programs\Python\Python311\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================================================== 3 passed, 1 warning in 0.42s ==========================================================================
```